Financial Markets

Master Study Notes

From first principles to active trading

Modules Covered

I. Introduction to Financial Markets
II. The Stock Market
III. Market Participants & Structure
IV. Fundamental Analysis
V. Technical Analysis
VI. Trading Mechanics
VII. Risk Management
VIII. Market Microstructure
IX. Macro & Global Markets
X. Advanced Topics & Next Steps

Personal notes — continuously updated

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Chapter 1

Introduction to Financial Markets

Note

This module builds the conceptual foundation. Every advanced topic later — options pricing, order flow, macro positioning — assumes fluency with what is covered here.

1.1 What Are Financial Markets?

A **financial market** is any organised or informal system in which buyers and sellers exchange financial instruments — assets that represent a claim on future cash flows, ownership, or contractual rights.

Markets exist to solve two universal problems:

1. **Capital allocation** — directing money from those who have it (savers, investors) to those who need it (companies, governments, individuals).
2. **Price discovery** — aggregating dispersed, private information from millions of participants into a single observable price.

Key Term: Financial Instrument

Any contract that creates a financial asset for one party and a financial liability (or equity) for another. Examples: shares, bonds, futures contracts, options, currencies, loans.

1.2 The Market Hierarchy

Market	What Trades Here
Equity Markets	Ownership stakes (shares/stocks) in companies
Debt / Bond Markets	Loans to governments and corporations
Money Markets	Very short-term debt (<1 year); cash equivalents
Foreign Exchange (FX)	Currency pairs
Commodity Markets	Physical goods: oil, gold, wheat, etc.
Derivatives Markets	Contracts derived from underlying assets (futures, options, swaps)
Crypto Markets	Digital/cryptographic assets

Note

These markets are deeply interconnected. A rising dollar ($) usually pressures commodity prices (they are dollar-denominated). Rising interest rates affect stock valuations via the discount rate. You will not understand equities fully without at minimum a passing knowledge of rates and FX.

1.3 Primary vs. Secondary Markets

Key Term: Primary Market

Where new securities are *created and sold for the first time*. A company doing an IPO (Initial Public Offering) is selling shares in the primary market. Proceeds go **to the issuer**.

Key Term: Secondary Market

Where previously issued securities are *traded between investors*. When you buy Apple stock on the NYSE, you are buying from another investor in the secondary market. Proceeds go **to the seller**, not to Apple.

This distinction matters: most people operate entirely in secondary markets, yet it is primary market activity (IPOs, bond issuance) that directly funds the real economy.

1.4 Why Prices Move

Price is determined by **supply and demand**. In financial markets, that comes down to:

- **Expectations** — prices are forward-looking. A stock represents the present value of all future cash flows, not current earnings alone.

- **New information** — earnings beats, central bank decisions, geopolitical shocks cause participants to re-assess expected future cash flows.
- **Sentiment and positioning** — markets overshoot fair value regularly because humans are emotional. Fear and greed are real forces.
- **Liquidity** — thin markets move more violently on the same information than deep ones.

Key Term: Efficient Market Hypothesis (EMH)

The theory that asset prices *fully reflect all available information* at any given moment, making it impossible to consistently outperform the market through skill alone. There are three forms: **weak** (prices reflect past prices), **semi-strong** (prices reflect all public information), and **strong** (prices reflect all information including insider knowledge). Widely debated — practical traders generally reject strong form.

1.5 Return and Risk — The Foundational Trade-off

$$\text{Return} = \frac{P_1 - P_0 + D}{P_0}$$

where P_0 = entry price, P_1 = exit price, D = dividends (or other income received).

Every additional unit of expected return requires accepting additional risk. This is the **risk-return trade-off** — the single most important idea in all of finance.

Common risk types:

- **Market risk** (systematic) — risk from broad market moves; cannot be diversified away.
- **Idiosyncratic risk** (unsystematic) — company-specific risk; can be reduced by diversification.
- **Liquidity risk** — inability to exit a position at the desired price.
- **Credit risk** — the counterparty may default.
- **Inflation risk** — real returns eroded by price level increases.
- **Tail risk** — low-probability, very high-impact events (black swans).

1.6 Time Value of Money

This is the mathematical backbone of valuation.

Future Value:

$$FV = PV \times (1 + r)^n$$

Present Value:

$$PV = \frac{FV}{(1 + r)^n}$$

where r = discount rate (required return), n = number of periods.

If you expect to receive \$1,000 in 3 years and your required return is 8%:

$$PV = \frac{1000}{(1.08)^3} = \frac{1000}{1.2597} \approx \$793.83$$

You should be willing to pay at most \$793.83 today for that future \$1,000.

1.7 Key Vocabulary to Memorise

Term	Definition
Bull market	Sustained price increases, typically >20% from trough
Bear market	Sustained price declines, typically >20% from peak
Correction	A 10–20% pullback from a recent high
Rally	A short-term price increase within any trend
Volatility	The degree of price fluctuation, often measured as standard deviation
Liquidity	How easily an asset can be bought or sold without moving the price
Spread	Difference between bid (buy) price and ask (sell) price
Long	Owning an asset; profit if price rises
Short	Borrowed and sold an asset; profit if price falls
Leverage	Using borrowed money to amplify position size
Margin	Collateral deposited to support a leveraged position
Mark-to-market	Daily revaluation of positions at current market prices

Leverage amplifies *both* gains and losses proportionally. A 2× leveraged position loses 20% when the underlying moves 10% against you. Many beginners blow up accounts by misunderstanding leverage.

Chapter 2

The Stock Market

2.1 What Is a Stock?

A **stock** (also called a **share** or **equity**) represents a unit of fractional ownership in a corporation. Shareholders are entitled to:

- A proportional claim on the company's **assets** (after all liabilities are settled)
- A proportional claim on future **dividends** (if declared)
- **Voting rights** on major corporate decisions (depending on share class)
- The ability to **sell** the share to someone else in the secondary market

Key Term: Common vs. Preferred Stock

Common stock — standard equity with voting rights; dividends are discretionary. **Preferred stock** — typically no voting rights but has a fixed, priority dividend claim before common shareholders. In bankruptcy, preferred ranks above common but below debt.

2.2 How Companies List on a Stock Exchange

2.2.1 Initial Public Offering (IPO)

A private company sells shares to the public for the first time. Process:

1. Company hires an **investment bank** (underwriter) to value the company and find buyers.
2. Underwriter sets an **offer price** and allocates shares to institutional investors.
3. Shares begin trading on the exchange — this is when retail investors typically get access.

2.2.2 Other Listing Methods

- **Direct Listing** — no new shares created; existing shareholders sell directly. No underwriter lockup. (Example: Spotify 2018)
- **SPAC** (Special Purpose Acquisition Company) — a shell company lists first, then merges with a private company.

2.3 Major Global Stock Exchanges

Exchange	Location	Notable Index
NYSE	New York, USA	Dow Jones Industrial Average
NASDAQ	New York, USA	NASDAQ Composite, NASDAQ 100
LSE	London, UK	FTSE 100
TSE	Tokyo, Japan	Nikkei 225
SSE	Shanghai, China	SSE Composite
Euronext	Amsterdam / Paris	CAC 40, AEX
SGX	Singapore	Straits Times Index (STI)

2.4 Stock Indices

An **index** is a mathematical basket of stocks used to represent the performance of a market segment or the whole market.

Key Term: Index Weighting

Price-weighted — higher-priced stocks have more influence (Dow Jones, Nikkei 225).
Market cap-weighted — stocks weighted by total market value; dominant method (S&P 500, FTSE 100).
Equal-weighted — each stock has the same weight regardless of size.

Market Capitalisation:

$$\text{Market Cap} = \text{Share Price} \times \text{Shares Outstanding}$$

2.4.1 The S&P 500 — The Benchmark That Matters Most

The S&P 500 tracks the 500 largest US publicly listed companies by market cap. It is the standard benchmark against which professional fund managers measure their performance. "Beating the market" typically means outperforming the S&P 500.

2.5 Sectors and Industries

The **GICS** (Global Industry Classification Standard) divides the market into 11 sectors:

Sector	Examples
Information Technology	Apple, Microsoft, NVIDIA
Health Care	Johnson & Johnson, Pfizer
Financials	JPMorgan, Goldman Sachs
Consumer Discretionary	Amazon, Tesla, Nike
Communication Services	Alphabet (Google), Meta
Industrials	Boeing, Caterpillar
Consumer Staples	Procter & Gamble, Coca-Cola
Energy	ExxonMobil, Chevron
Utilities	NextEra Energy
Real Estate	Simon Property Group, REITs
Materials	Linde, Freeport-McMoRan

Note

Sectors rotate in and out of favour with the economic cycle. Defensive sectors (Staples, Utilities, Health Care) hold up better in downturns. Cyclical sectors (Technology, Discretionary, Industrials) outperform in expansions. This is called **sector rotation**.

2.6 How Stock Prices Are Determined — A First Pass

Short-term prices are set by **supply and demand** in the order book. Long-term prices converge toward **fundamental value** — the present value of future cash flows.

The simplest valuation model is the **Gordon Growth Model** (for a dividend-paying stock):

$$P_0 = \frac{D_1}{r - g}$$

where D_1 = expected dividend next year, r = required rate of return, g = sustainable dividend growth rate.

A stock pays \$2 dividend, expected to grow at 4% forever. Investors require 9%

return.

$$P_0 = \frac{2}{0.09 - 0.04} = \frac{2}{0.05} = \$40$$

If the price rises above \$40, the stock is overvalued relative to this model.

2.7 Corporate Actions

Events a company can take that affect its share price and structure:

Action	Effect
Dividend	Cash paid to shareholders; share price falls by approximately the dividend amount on ex-date
Stock Split	More shares at lower price; no change to market cap (e.g., 2-for-1 split)
Reverse Split	Fewer shares at higher price; often signals distress
Share Buyback	Company purchases its own shares; reduces float, typically boosts EPS
Rights Issue	New shares sold to existing holders at a discount; dilutes existing shareholders
M&A	Merger or acquisition; target often trades up to bid price
Spin-off	Division separated into independent public company

Chapter 3

Market Participants & Structure

Note

Understanding *who* is in the market and *why* they trade changes everything. Price action is the aggregate footprint of all participants acting on their different objectives, time horizons, and information sets.

3.1 The Buy Side vs. The Sell Side

Key Term: Sell Side

Firms that *create and sell* financial products, execute client trades, and provide research. They earn through commissions, spreads, and underwriting fees.
Examples: Goldman Sachs, Morgan Stanley, JPMorgan (as investment banks), market makers.

Key Term: Buy Side

Firms that *buy* securities with the goal of generating returns for themselves or clients.
Examples: mutual funds, hedge funds, pension funds, insurance companies, endowments.

3.2 Types of Market Participants

3.2.1 Retail Investors

Individual non-professional investors. Trade through brokers (Fidelity, Schwab, Interactive Brokers, Robinhood). Generally small position sizes. Collectively significant but individually price-takers.

3.2.2 Institutional Investors

Manage large pools of capital:

- **Mutual Funds** — pool retail money, must disclose holdings quarterly, cannot short.
- **Pension Funds** — very long-duration mandates; large in bonds and equities.
- **Insurance Companies** — must match asset duration to liability duration; bond-heavy.
- **Endowments** — universities, foundations; long-horizon, often hold alternatives.
- **Sovereign Wealth Funds** — government-owned, very large (e.g., Norway GPFG, GIC).

3.2.3 Hedge Funds

Lightly regulated investment vehicles for sophisticated investors. Can use: leverage, short selling, derivatives, concentrated positions. Performance fee structure: typically **2 and 20** (2% management fee + 20% of profits).

Strategies include: long/short equity, global macro, statistical arbitrage, event-driven, distressed credit, systematic/quant.

3.2.4 Market Makers

Provide continuous two-sided quotes (bid and ask). Profit from the **bid-ask spread**. Obligated to buy and sell; absorb order imbalances. Essential for market liquidity. Examples: Citadel Securities, Virtu Financial, Jane Street.

3.2.5 High-Frequency Traders (HFT)

Use ultra-fast execution (microseconds) and algorithmic strategies to profit from tiny, fleeting price inefficiencies. Controversial — seen by some as liquidity providers, by others as front-runners.

3.2.6 Central Banks

The Federal Reserve (US), ECB (Europe), BOJ (Japan) can buy and sell assets directly. Quantitative easing (QE) involves large-scale asset purchases, injecting liquidity. Central bank actions have outsized market impact.

3.3 How a Trade Is Executed

The lifecycle of a typical equity trade:

Step 1. **Order Entry** — Investor places order via broker (market, limit, stop, etc.)

Step 2. **Routing** — Broker routes to exchange, ATS (Alternative Trading System), or internalises against market maker flow.

Step 3. **Matching** — Order matching engine pairs buy and sell orders at a price.

Step 4. **Execution** — Trade confirmed. Price and quantity locked in.

Step 5. **Clearing** — Clearinghouse (e.g., DTCC in the US) becomes the counterparty to both sides, managing counterparty risk.

Step 6. **Settlement** — Actual transfer of securities and cash. US equities: **T+1** (trade date plus one business day).

3.4 Order Types

Order Type	Description
Market Order	Execute immediately at best available price. Fast but no price guarantee.
Limit Order	Execute only at a specified price or better. Price control but may not fill.
Stop Order	Becomes a market order once a trigger price is hit. Used to cut losses.
Stop-Limit	Becomes a *limit* order once trigger is hit. Risk of non-fill in fast markets.
Trailing Stop	Stop that moves with the price, locking in gains.
Iceberg Order	Only a small portion of total size is visible; remainder is hidden.
Fill or Kill	Execute the entire order immediately or cancel it entirely.

In fast-moving, illiquid, or gap-prone markets, **stop orders** can fill significantly worse than the trigger price (*slippage*). Never assume a stop order equals a guaranteed exit price.

3.5 Brokers and Execution Venues

3.5.1 Types of Brokers

- **Full-service broker** — advice, research, wealth management. Higher fees.
- **Discount broker** — execution only, low/zero commission. (Schwab, Fidelity, IBKR)
- **Direct Access Broker** — routes directly to exchanges; used by active traders and HFT. (Interactive Brokers Pro, Lightspeed, DAS Trader)

3.5.2 Payment for Order Flow (PFOF)

Retail brokers like Robinhood route orders to market makers (e.g., Citadel Securities) in exchange for payment. Controversial: critics argue retail gets slightly worse execution; defenders say it funds zero-commission trading. **Banned in the UK and EU.**

3.5.3 Dark Pools

Private exchanges where large institutional orders are matched anonymously, away from public markets, to minimise **market impact**. A large sell order on a public exchange signals intent and can move the price adversely before execution completes.

Chapter 4

Fundamental Analysis

Note

Fundamental analysis asks: **what is this asset actually worth?** It grounds investment decisions in business reality rather than price patterns alone. Warren Buffett, Charlie Munger, and most long-term investors use this approach.

4.1 The Three Financial Statements

Every listed company publishes three core statements. Learn to read all three.

4.1.1 Income Statement (Profit & Loss)

Shows **revenues, expenses, and profit** over a period (quarterly or annually).

Line Item	What It Means
Revenue / Sales	Total money earned from operations
Cost of Goods Sold (COGS)	Direct costs of producing goods/services
Gross Profit	Revenue − COGS
Operating Expenses (OpEx)	R&D, SG&A, depreciation
EBIT	Earnings Before Interest and Tax
EBITDA	EBIT + Depreciation + Amortisation; proxy for operating cash
Net Income	Final profit after interest and tax
EPS	Earnings Per Share = Net Income ÷ Shares Outstanding

4.1.2 Balance Sheet

A **snapshot** of what the company owns (assets) and owes (liabilities) at one moment.

$$\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}$$

This equation always holds. Equity is the residual claim.

4.1.3 Cash Flow Statement

Tracks **actual cash movement**, split into three activities:

- **Operating Cash Flow (OCF)** — cash from core business. Most important signal of health.
- **Investing Cash Flow** — capex, acquisitions, asset sales.
- **Financing Cash Flow** — debt issuance/repayment, dividends, share buybacks.

A company can show positive net income while *burning cash*. Always check the cash flow statement. Enron's fraud persisted partly because investors ignored cash flow signals.

4.2 Key Financial Ratios

4.2.1 Valuation Ratios

Ratio	Formula	What It Tells You
P/E	Price ÷ EPS	How much investors pay per $1 of earnings
P/S	Market Cap ÷ Revenue	Useful for unprofitable companies
P/B	Price ÷ Book Value per Share	Premium over net assets
EV/EBITDA	Enterprise Value ÷ EBITDA	Capital-structure-neutral valuation
PEG	P/E ÷ EPS Growth Rate	Adjusts P/E for growth

Enterprise Value (EV):

$$EV = \text{Market Cap} + \text{Total Debt} - \text{Cash \& Equivalents}$$

EV represents the *total cost to acquire* a business.

4.2.2 Profitability Ratios

Ratio	Formula	Interpretation
Gross Margin	Gross Profit ÷ Revenue	Pricing power, cost efficiency
Operating Margin	EBIT ÷ Revenue	Core business profitability
Net Margin	Net Income ÷ Revenue	Bottom-line efficiency
ROE	Net Income ÷ Equity	Return on shareholders' capital
ROA	Net Income ÷ Total Assets	Asset efficiency
ROIC	NOPAT ÷ Invested Capital	True economic return on capital deployed

4.2.3 Liquidity and Leverage Ratios

Ratio	Formula	Interpretation
Current Ratio	Current Assets ÷ Current Liabilities	Short-term solvency
Quick Ratio	(Current Assets − Inventory) ÷ CL	Stricter solvency test
Debt-to-Equity	Total Debt ÷ Equity	Financial leverage
Net Debt/EBITDA	Net Debt ÷ EBITDA	Years to repay debt from operations
Interest Coverage	EBIT ÷ Interest Expense	Ability to service debt

4.3 Discounted Cash Flow (DCF) Valuation

The theoretical gold standard: value a company as the present value of all future free cash flows (FCF), discounted at the **Weighted Average Cost of Capital (WACC)**.

$$\text{Intrinsic Value} = \sum_{t=1}^{n} \frac{FCF_t}{(1+WACC)^t} + \frac{TV}{(1+WACC)^n}$$

where TV = terminal value (value of all cash flows beyond the forecast period).
Free Cash Flow:

$$FCF = \text{Operating Cash Flow} - \text{Capital Expenditure}$$

WACC:

$$WACC = \frac{E}{V} \cdot r_e + \frac{D}{V} \cdot r_d \cdot (1-T)$$

where E/V = equity weight, D/V = debt weight, r_e = cost of equity, r_d = cost of debt, T = tax rate.

Note

DCF is highly sensitive to assumptions — a 1% change in WACC or growth rate can change the output by 30–50%. Use it as a *range*, not a precise number. "Garbage in, garbage out."

4.4 Qualitative Factors

Numbers are necessary but not sufficient. Also assess:

- **Competitive moat** — durable competitive advantage (brand, network effects, switching costs, cost advantages, intangible assets). Popularised by Warren Buffett.
- **Management quality** — capital allocation history, insider ownership, communication.
- **Industry dynamics** — growth trajectory, competitive intensity (Porter's Five Forces).
- **Regulatory risk** — antitrust, environmental, financial regulation exposure.
- **Balance sheet resilience** — can the company survive a recession or industry shock?

4.5 Reading Earnings Reports

Every quarter, public companies release:

1. **Earnings Release** — headline EPS, revenue, and guidance.
2. **10-Q / 10-K** (US) — detailed SEC filings with full financial statements.
3. **Earnings Call** — management discusses results and takes analyst questions.

Note

Markets react to **surprises vs. expectations**, not absolute numbers. A company with great results can fall in price if the results missed consensus estimates. Track **EPS surprise** and **revenue surprise** alongside guidance changes.

Chapter 5

Technical Analysis

Note

Technical analysis (TA) is the study of *price and volume history* to forecast future price direction. It operates on the premise that all known information is reflected in the price, and that crowd psychology creates recurring, identifiable patterns.

5.1 Core Assumptions of Technical Analysis

1. **Price discounts everything** — market price reflects all fundamentals, sentiment, and information. Analysis focuses on price itself.
2. **Prices move in trends** — once established, a trend tends to continue until it reverses. The trend is your friend.
3. **History repeats** — patterns recur because human psychology is consistent. Fear and greed create the same structures again and again.

5.2 Chart Types

5.2.1 Candlestick Charts

The most common chart type. Each candle represents a time period (1min, 1hr, 1day, etc.)

Candle Part	Meaning
Body	Range between Open and Close price
Upper Wick	Highest price reached during the period
Lower Wick	Lowest price reached during the period
Green / White body	Close > Open (bullish period)
Red / Black body	Close < Open (bearish period)

5.2.2 Key Candlestick Patterns

Pattern	Signal	Description
Doji	Indecision	Open ≈ Close; wicks on both sides
Hammer	Bullish reversal	Small body at top, long lower wick
Shooting Star	Bearish reversal	Small body at bottom, long upper wick
Engulfing	Reversal	Second candle fully engulfs first
Marubozu	Strong trend	Full body, no wicks

5.3 Trend Analysis

5.3.1 Identifying Trends

- **Uptrend** — series of Higher Highs (HH) and Higher Lows (HL)
- **Downtrend** — series of Lower Highs (LH) and Lower Lows (LL)
- **Sideways / Ranging** — price oscillates between a ceiling and a floor

5.3.2 Trendlines

A line connecting a series of swing lows in an uptrend (or swing highs in a downtrend). A break of a trendline is a potential signal of trend reversal.

5.3.3 Support and Resistance

Key Term: Support

A price level where buying interest is strong enough to prevent further decline. Price has bounced from this level before. It is psychological "floor".

Key Term: Resistance

A price level where selling pressure prevents further advance. Price has been rejected here before. Acts as a psychological "ceiling".

Note

Role Reversal: Once broken, a support level often becomes new resistance, and a resistance level often becomes new support. This is one of the most reliable observations in all of technical analysis.

5.4 Moving Averages

A moving average smooths price data to reveal the underlying trend direction.

Simple Moving Average (SMA):

$$SMA_n = \frac{P_1 + P_2 + \cdots + P_n}{n}$$

Exponential Moving Average (EMA):

$$EMA_t = P_t \cdot k + EMA_{t-1} \cdot (1 - k), \quad k = \frac{2}{n+1}$$

EMA gives more weight to recent prices; reacts faster than SMA.

5.4.1 Common MA Periods

- **20 / 21 EMA** — short-term trend; used by swing traders
- **50 SMA** — medium-term trend; key institutional reference
- **200 SMA** — long-term trend; major bull/bear dividing line

5.4.2 Moving Average Crossovers

- **Golden Cross** — 50 SMA crosses above 200 SMA. Bullish signal.
- **Death Cross** — 50 SMA crosses below 200 SMA. Bearish signal.

5.5 Key Technical Indicators

5.5.1 Relative Strength Index (RSI)

Measures the speed and magnitude of price changes. Oscillates between 0 and 100.

$$RSI = 100 - \frac{100}{1 + RS}, \quad RS = \frac{\text{Average Gain (14 periods)}}{\text{Average Loss (14 periods)}}$$

- RSI > 70: Overbought (potential sell signal)
- RSI < 30: Oversold (potential buy signal)
- Divergence: Price makes new high but RSI does not — bearish divergence.

5.5.2 MACD (Moving Average Convergence Divergence)

Two EMAs (12 and 26 periods) and a signal line (9-period EMA of MACD).

- **MACD crosses above signal line** — bullish
- **MACD crosses below signal line** — bearish
- **Histogram** — visualises the gap between MACD and signal line

5.5.3 Bollinger Bands

A 20-period SMA with bands ±2 standard deviations above and below.

- Price touching upper band: potentially overbought or strong trend continuation
- Price touching lower band: potentially oversold
- **Squeeze**: narrow bands signal low volatility — often precedes a big move

5.5.4 Volume

Volume validates price moves. High volume on a breakout = more convincing. Low volume on a rally = suspect. Watch for **volume divergence**.

5.6 Chart Patterns

5.6.1 Continuation Patterns

- **Flag / Pennant** — tight consolidation after a sharp move; usually resolves in trend direction
- **Triangle** (ascending, descending, symmetrical) — price coiling; breakout expected
- **Cup and Handle** — U-shaped recovery, small pullback, then breakout

5.6.2 Reversal Patterns

- **Head and Shoulders** — three peaks; middle (head) is highest; neckline break = sell signal
- **Inverse Head and Shoulders** — same structure inverted; neckline break = buy signal
- **Double Top / Double Bottom** — price fails twice at same level; signals reversal

5.7 Fibonacci Retracements

Based on the Fibonacci sequence, key retracement levels are: **23.6%, 38.2%, 50%, 61.8%, 78.6%**

The 61.8% level is the "Golden Ratio" and is considered the most significant. Traders use these levels as potential support/resistance zones after a move.

Technical analysis is descriptive, not prescriptive. No indicator works all the time. Use TA as a *framework for probability*, not as a crystal ball. Always combine with risk management; patterns fail regularly.

Chapter 6

Trading Mechanics

6.1 Trading Timeframes

Different traders operate on completely different timescales with different objectives:

Style	Hold Period	Approach
Scalping	Seconds – Minutes	Tiny margins, massive volume; low retail suitability
Day Trading	Intraday only	No overnight positions; requires intense focus
Swing Trading	Days – Weeks	Catch medium moves; most accessible to part-time traders
Position Trading	Weeks – Months	Trend-following; fewer decisions required
Investing	Months – Years	Fundamentals-driven; buy and hold

Note

Most beginners should start with **swing trading or position trading**. Day trading and scalping require exceptional skill, fast infrastructure, and very low commissions to be profitable. Studies consistently show >70% of day traders lose money.

6.2 The Trading Plan

Every trade should follow a pre-defined plan. Never trade without one.

A complete trade plan defines:

1. **Setup / Thesis** — why does this trade make sense? What is the edge?
2. **Entry criteria** — exact conditions that trigger entry (price level, indicator, event)
3. **Stop loss** — where are you wrong? Maximum loss you accept on this trade.
4. **Target(s)** — where do you take profit? Minimum first target.

5. **Position size** — how many shares/contracts, given your stop and risk per trade.
6. **Trade management** — do you scale out? Trail a stop? Let it run?
7. **Invalidation** — what price/event would invalidate the entire thesis?

6.3 Position Sizing

This is how much capital you allocate to a single trade. Critical for survival.

Risk-based position sizing:

$$\text{Position Size} = \frac{\text{Account Risk (\$)}}{\text{Stop Distance per Share (\$)}}$$

Account Risk:

$$\text{Account Risk} = \text{Account Size} \times \text{Risk \% per Trade}$$

Account: \$20,000. Risk 1% per trade. Entry: \$50. Stop: \$47.

$$\text{Account Risk} = \$20{,}000 \times 0.01 = \$200$$

$$\text{Stop Distance} = \$50 - \$47 = \$3$$

$$\text{Position Size} = \frac{\$200}{\$3} = 66 \text{ shares}$$

Maximum loss on the trade = \$200, regardless of position size.

Common risk per trade for beginners: 0.5%–1%. Professional traders often risk 0.25%–0.5% per trade.

6.4 Risk-Reward Ratio

Key Term: Risk-Reward Ratio (RRR)

The ratio of potential profit to potential loss on a trade.

$$RRR = \frac{\text{Target Distance}}{\text{Stop Distance}}$$

Entry: \$50. Stop: \$47 (risk = \$3). Target: \$59 (reward = \$9).
RRR = 9/3 = **3:1**

With a 3:1 RRR, you only need to be right **25% of the time** to break even. Most professional traders target a minimum of **2:1 or 3:1** RRR.

Breakeven Win Rate:

$$W_{\text{min}} = \frac{1}{1 + RRR}$$

At 2:1 RRR: $W_{\text{min}} = 1/3 \approx 33\%$. At 3:1: $W_{\text{min}} = 25\%$.

6.5 Expected Value (EV)

The single most important concept for evaluating a trading strategy:

$$EV = (P_{\text{win}} \times \text{Avg Win}) - (P_{\text{loss}} \times \text{Avg Loss})$$

Win rate = 40%. Average win = \$300. Average loss = \$100.

$$EV = (0.40 \times 300) - (0.60 \times 100) = 120 - 60 = +\$60 \text{ per trade}$$

This is a positive expectancy system — profitable over many repetitions.

6.6 Long vs. Short Selling

6.6.1 Going Long

Buy shares, hold, sell later at higher price. Maximum loss = 100% of capital invested. Unlimited upside.

6.6.2 Short Selling

1. **Borrow** shares from broker (pays a borrow fee).
2. **Sell** those shares at current market price.
3. **Buy back** ("cover") at a lower price later.
4. **Return** shares to lender and keep the difference.

Short selling has **theoretically unlimited risk** — a stock can rise to infinity. Losses are not capped. Additionally, a **short squeeze** can occur when many short sellers are forced to buy simultaneously, catapulting the price rapidly (e.g., GameStop 2021).

6.7 Leverage and Margin

6.7.1 How Margin Works

Margin allows you to borrow from your broker to control a larger position. A **2:1 margin** means you can control $20,000 of assets with $10,000 of capital.

6.7.2 Margin Call

If losses erode your equity below the broker's **maintenance margin requirement**, you receive a margin call: deposit more funds or positions are forcibly liquidated — often at the worst possible time.

Leverage is not a tool for beginners. Master position sizing and risk management with *no leverage* first. Many traders blow up accounts by using leverage before they have developed the discipline and edge to use it responsibly.

6.8 Trading Psychology

Markets are 50% mechanics and 50% psychology. Common destructive patterns:

- **Revenge trading** — taking impulsive trades to recover a loss immediately. Almost always leads to larger losses.
- **FOMO** (Fear of Missing Out) — chasing price after missing an entry.
- **Letting losers run** — not respecting stop losses, hoping it will come back.
- **Cutting winners short** — exiting too early out of fear of giving back gains.
- **Overtrading** — taking too many trades; dilutes edge and increases commission costs.
- **Confirmation bias** — seeking information that confirms your position, ignoring contradictory evidence.

Note

A losing trade *executed according to plan* is a good trade. A winning trade taken outside your rules is a bad trade. **Process over outcome.** Trading is a probability game measured over hundreds of trades, not individual outcomes.

Chapter 7

Risk Management

Note

Risk management is what keeps you in the game. Every professional trader's primary goal is capital preservation — the ability to keep playing. Edge compounds over time only if you survive the drawdowns. The best entry signal in the world is worthless without a disciplined risk framework.

7.1 Drawdown

Key Term: Drawdown (DD)

The peak-to-trough decline in portfolio value, expressed as a percentage.

$$DD = \frac{P_{\text{peak}} - P_{\text{trough}}}{P_{\text{peak}}} \times 100\%$$

A 50% drawdown requires a **100% return** just to get back to breakeven.

Loss	Recovery Required	Implied Effect
10%	11.1%	Manageable
20%	25.0%	Uncomfortable
30%	42.9%	Serious
40%	66.7%	Very serious
50%	100.0%	Account in crisis
75%	300.0%	Near impossible

This is why limiting drawdowns is more important than maximising returns.

7.2 Portfolio-Level Risk Rules

7.2.1 Daily / Weekly Loss Limits

Set a **circuit breaker**: if you lose $X\%$ in a day (e.g., 2%), stop trading. Come back tomorrow. This prevents emotional spiralling from turning a bad day into a catastrophe.

7.2.2 Maximum Open Risk

At any given time, total open risk across all positions should be capped. Example: no more than 5–6% of account at risk simultaneously.

7.2.3 Correlation Risk

Holding 10 positions in the same sector = effectively one concentrated bet. True diversification requires **uncorrelated** positions.

7.3 Value at Risk (VaR)

Key Term: Value at Risk (VaR)

The maximum expected loss over a given time horizon at a specified confidence level. "95% 1-day VaR of $10,000" means: there is a 5% chance of losing more than $10,000 in one day.

VaR fails precisely when you need it most — in fat-tail events and market crises, correlations spike to 1 and losses exceed VaR estimates dramatically. The 2008 financial crisis exposed VaR's limitations. Use it as one tool, not the only tool.

7.4 The Kelly Criterion

Optimal position sizing formula to maximise long-run portfolio growth.

$$f^* = \frac{p}{a} - \frac{q}{b}$$

where:

- f^* = fraction of capital to risk
- p = probability of winning
- q = probability of losing ($= 1 - p$)
- b = net profit per unit risked (win/loss ratio)
- a = loss per unit risked ($= 1$ in standard form)

Win rate = 55%, Win/Loss ratio = 1.5.

$$f^* = \frac{0.55}{1} - \frac{0.45}{1.5} = 0.55 - 0.30 = 0.25 = 25\%$$

Full Kelly says bet 25% of capital. In practice, use **Half Kelly** (12.5%) to reduce volatility while capturing most of the growth benefit.

Note

Kelly sizing assumes you know your true edge precisely — rarely the case in markets. Over-betting Kelly leads to catastrophic drawdowns. Most professional traders use a fraction of Kelly (25–50%).

7.5 Sharpe Ratio and Performance Metrics

Sharpe Ratio:

$$S = \frac{R_p - R_f}{\sigma_p}$$

where R_p = portfolio return, R_f = risk-free rate, σ_p = standard deviation of returns.

A Sharpe Ratio of 1.0 is decent; 2.0 is very good; 3.0+ is exceptional (most quant funds target this).

Metric	What It Measures
Sharpe Ratio	Return per unit of total volatility
Sortino Ratio	Like Sharpe but only penalises *downside* volatility
Calmar Ratio	Annualised return ÷ Max Drawdown
Win Rate	% of trades that are profitable
Profit Factor	Gross profit ÷ Gross loss (want > 1.5)
Expectancy	Average $ made per trade (positive = edge)

7.6 Hedging

Reducing existing risk by taking an offsetting position.

- **Long stock + Buy Put option** — limits downside loss on the stock position.
- **Long stock + Short futures** — locks in a price; used by producers and large funds.
- **Long equities + Long VIX calls** — hedge against volatility spikes in crises.

- **Pair trade** — long one stock, short a correlated stock in same sector. Removes market beta; profits from relative performance.

Hedges have a cost (premium, drag, opportunity cost). Perfect hedges eliminate profit as well as loss. In practice, **partial hedges** are used.

7.7 The Trader's Ruin Problem

Even with a positive edge, a trader can go bankrupt through bad luck and poor sizing.

Probability of Ruin (simplified):

$$P(\text{ruin}) = \left(\frac{q}{p}\right)^N$$

where N = number of units of risk in the account, p = win probability, q = loss probability.

The message: **adequate capitalisation** combined with **small risk per trade** dramatically reduces the probability of ruin, even in a random walk with slight positive edge.

Never risk money you cannot afford to lose. Do not trade with rent money, emergency funds, or borrowed capital. The psychological pressure of trading money you *need* destroys decision-making and guarantees losses.

Chapter 8

Market Microstructure

Note

Market microstructure is the study of how markets actually function at a mechanical level — order books, liquidity, price formation, and the behaviour of market participants at fine-grained timescales. Understanding this gives traders an edge in execution and helps explain "why prices do what they do."

8.1 The Order Book

The **order book** (Level 2 data) is a real-time record of all outstanding buy and sell limit orders at each price level.

Bid (Buy) Size	**Price**	Ask (Sell) Size
500	**99.97**	
1200	**99.98**	
3000	**99.99**	
	— SPREAD —	
	100.01	2500
	100.02	1100
	100.03	800

- **Bid** — highest price a buyer is willing to pay ($99.99)
- **Ask** (Offer) — lowest price a seller is willing to accept ($100.01)
- **Spread** — difference = $0.02 in this case
- **Mid price** — (Bid + Ask) / 2 = $100.00
- **Depth** — the size available at each price level

8.2 Liquidity and Market Impact

Key Term: Liquidity

The ease with which a security can be bought or sold without meaningfully affecting its price. Measured by: spread (tighter = more liquid), depth (more size = more liquid), and market impact (large orders moving the price).

A **market order** eats through the order book. A large market buy order "walks up the book", consuming all available ask orders until filled — each successive fill occurs at a higher price. This is **market impact / slippage**.

Note

For small retail traders, slippage is negligible on liquid large-cap stocks. For institutional traders executing millions of shares, market impact is a major cost. This is why institutions use algorithmic execution (VWAP, TWAP, Implementation Shortfall).

8.3 Bid-Ask Spread as a Cost

Every time you cross the spread, you pay a cost:

$$\text{Spread Cost (round trip)} = \frac{\text{Ask} - \text{Bid}}{\text{Mid}} \times 2 \times \text{Position Value}$$

For a stock with a \$0.05 spread at \$50 mid, round-trip spread cost = 0.2%. At \$100,000 position size: \$200 per round trip — before any other cost.

8.4 Types of Orders and the Book

- **Limit order** — added to the book; provides liquidity. Often earns a **maker rebate** on exchanges.
- **Market order** — removes from the book; takes liquidity. Pays a **taker fee**.
- **Iceberg order** — large order with only part visible; hides intent.
- **Stop order** — hidden from the book; triggered and becomes market order at stop price.

8.5 Price Discovery

Key Term: Price Discovery

The process by which markets aggregate information from all participants into a single observed price. More informative prices = more efficient allocation of capital.

Price discovery happens continuously during market hours. **Opening auctions** and **closing auctions** are special mechanisms that batch orders to set the open and close price — the most liquid moments of the trading day.

8.5.1 Pre-Market and After-Hours Trading

Trading outside regular hours (US: 9:30am–4:00pm ET). Characteristics:

- Much lower liquidity and volume
- Wider spreads
- More volatile price action
- Earnings releases often happen after hours or pre-market

8.6 Volume Analysis

Volume is the number of shares/contracts traded in a period. It is the **second most important data point after price**.

- **High volume breakout** — conviction; more participants validating the move.
- **Low volume breakout** — suspect; may fail or reverse.
- **Climax volume** — exhaustion spike; potential reversal signal.
- **Volume dry-up** — lack of sellers (in uptrend) or buyers (in downtrend); often precedes continuation.

8.6.1 VWAP (Volume-Weighted Average Price)

$$VWAP = \frac{\sum(Price_i \times Volume_i)}{\sum Volume_i}$$

VWAP is the average price weighted by volume. Institutional traders use it as a benchmark. Trading above VWAP = bullish intraday; below = bearish intraday.

8.7 Market Regimes and Volatility

8.7.1 VIX — The Fear Index

The **CBOE Volatility Index (VIX)** measures the market's expectation of 30-day S&P 500 volatility, derived from options pricing.

- VIX < 15: Low volatility, complacent market
- VIX 15–25: Normal market
- VIX 25–40: Elevated fear
- VIX > 40: Crisis, extreme fear (March 2020 hit 85)

Note

High VIX environments have **wider spreads**, **faster moves**, and **larger gaps**. Position sizes should be *reduced* in high volatility regimes to maintain constant dollar risk.

8.8 Order Flow and Tape Reading

Order flow refers to the *real-time stream of transactions* executed in the market. Advanced traders read the tape (time and sales) to detect:

- Aggressive buying (large market orders lifting the ask)
- Aggressive selling (large orders hitting the bid)
- Absorption (large hidden orders absorbing flow without price movement)
- **Stop runs** — price briefly moves to known stop loss zones, triggers stops, then reverses. Common manipulation tactic by larger players.

Order flow analysis has a steep learning curve. Do not attempt it until you have mastered basic chart reading, risk management, and have consistent results on higher timeframes first.

Chapter 9

Macro & Global Markets

Note

No asset exists in isolation. Stock prices, bond yields, exchange rates, and commodity prices are connected through **macroeconomic forces**. Understanding the macro backdrop is essential for context — even if you trade purely technically.

9.1 The Economic Cycle

The economy moves in cycles: **expansion** → **peak** → **contraction/recession** → **trough** → expansion again.

9.1.1 Sector Performance Across the Cycle

Cycle Phase	Economy Characteristics	Outperforming Sectors
Early Expansion	Recovery, low rates, rising confidence	Consumer Discretionary, Financials
Mid Expansion	Solid growth, moderate inflation	Technology, Industrials, Materials
Late Expansion	High inflation, rising rates	Energy, Commodities, Real Estate
Recession	Contraction, falling rates	Consumer Staples, Utilities, Health Care

9.2 Interest Rates and the Bond Market

9.2.1 Central Banks and the Policy Rate

The central bank sets a **short-term policy interest rate** (e.g., the Federal Funds Rate). This rate propagates through the entire financial system:

- **Higher rates** → borrowing more expensive → lower business investment, lower consumer spending → slows inflation but also growth
- **Lower rates** → borrowing cheaper → more investment, more consumption → stimulates growth but risks inflation

9.2.2 The Yield Curve

A plot of **bond yields** (interest rates) across maturities (3m, 1y, 2y, 5y, 10y, 30y).

Key Term: Yield Curve Inversion

When short-term yields exceed long-term yields. Historically, an inverted yield curve has preceded every US recession since the 1950s. The 2-year vs 10-year spread is most watched.

9.2.3 Bond Prices and Yields — Inverse Relationship

$$\text{Bond Price} \propto \frac{1}{\text{Yield}}$$

When yields rise, existing bond prices fall. When yields fall, bond prices rise. This is because the fixed coupon payment becomes relatively less/more attractive compared to new bonds.

9.2.4 How Rates Affect Stock Valuations

Higher interest rates:

1. Raise the discount rate in DCF models → lower present value → lower stock prices.
2. Make risk-free returns (bonds, cash) more competitive vs. stocks.
3. Increase corporate borrowing costs → compress earnings.

Growth stocks (with cash flows far in the future) are especially rate-sensitive.

9.3 Inflation

Key Term: Inflation

The rate at which the general level of prices for goods and services rises, eroding purchasing power. Measured by CPI (Consumer Price Index) and PCE (Personal Consumption Expenditures).

9.3.1 Inflation's Effect on Markets

- **Moderate inflation** (2–3%): Healthy; supports nominal earnings growth.

- **High inflation**: Central banks raise rates → negative for bonds and growth stocks; commodities and energy often outperform.
- **Deflation**: Very negative; reduces corporate revenues; Japan's "Lost Decades".

9.4 Key Macro Economic Indicators

Indicator	Release Frequency	What Traders Watch
Non-Farm Payrolls (NFP)	Monthly (US)	Employment health; market-moving
CPI / Core CPI	Monthly	Inflation trajectory; Fed direction
GDP Growth	Quarterly	Economic expansion/contraction
PMI (Manufacturing/Services)	Monthly	Real-time business activity
Retail Sales	Monthly	Consumer spending health
Jobless Claims	Weekly	Leading employment indicator
Fed Funds Rate Decision	8x/year	Defines cost of money
FOMC Minutes	8x/year	Fed deliberation detail

Major macro releases (**NFP**, **CPI**, **FOMC**) cause violent, fast price moves with wide spreads. Experienced traders often close or hedge positions before these releases. Never hold leveraged positions through major macro events unless it is part of a deliberate, sized strategy.

9.5 Currency Markets (FX)

Exchange rates affect:

- **Multinational earnings** — a strong USD reduces the dollar value of foreign revenues.
- **Commodity prices** — most commodities are priced in USD; strong dollar → lower prices.
- **Emerging markets** — EM countries with USD-denominated debt suffer when USD strengthens.

9.5.1 Key Currency Relationships

- **USD/JPY** — risk barometer; yen strengthens in risk-off environments
- **AUD/USD, CAD/USD** — commodity currencies; follow commodity prices
- **EUR/USD** — most liquid FX pair; follows ECB vs Fed policy divergence

- **DXY** — US Dollar Index; inverse relationship with most risk assets

9.6 Commodities

Commodity	Category	Key Drivers
Crude Oil (WTI/Brent)	Energy	OPEC policy, global demand, USD
Gold	Precious Metal	Real interest rates, USD, safe haven demand
Silver	Precious/Industrial	Gold correlation + industrial demand
Copper	Industrial Metal	China demand; economic growth proxy
Wheat/Corn/Soy	Agriculturals	Weather, geopolitics, USD

Gold and real interest rates have a strong inverse relationship:

$$\text{Real Rate} = \text{Nominal Rate} - \text{Expected Inflation}$$

When real rates fall (e.g., negative real rates), gold becomes more attractive as a store of value vs. bonds.

9.7 Inter-Market Analysis

The four major asset classes — equities, bonds, commodities, and currencies — are linked. Classic relationships (not always reliable):

- Bonds ↑ (yields ↓) → stocks ↑ (historically)
- Dollar ↑ → gold ↓, commodities ↓, EM ↓
- Commodities ↑ → inflation expectations ↑ → bonds ↓
- Copper ↑ → global growth optimism → cyclical stocks ↑

Note

These correlations **shift through regimes**. The bond-equity correlation was negative for 20 years (2000–2020) and turned positive in the 2022 inflation shock. Always verify whether historical relationships still hold in the current regime.

Chapter 10

Advanced Topics & Next Steps

Note

This module introduces areas to explore as you develop mastery of the foundations. Each topic below is a deep discipline in its own right. Treat these as signposts for your learning roadmap, not immediate requirements.

10.1 Options — The Basics

An **option** is a contract giving the holder the **right, but not the obligation**, to buy or sell an asset at a specified **strike price** by or on the **expiration date**.

Type	Position	Rights and Risk
Call	**Long**	Right to *buy* at strike. Profit if price rises above strike + premium.
Call	**Short**	Obligation to *sell* at strike. Limited profit (premium), unlimited risk.
Put	**Long**	Right to *sell* at strike. Profit if price falls below strike + premium.
Put	**Short**	Obligation to *buy* at strike. Limited profit (premium), large downside.

10.1.1 The Greeks

Options price sensitivity measures:

Greek	What It Measures
Delta (Δ)	Change in option price per \$1 change in underlying. Ranges 0–1 (calls), 0 to -1 (puts).
Gamma (Γ)	Rate of change of Delta. High near expiration.
Theta (Θ)	Time decay. Options lose value as expiration approaches.
Vega ($\mathcal{V}$)	Sensitivity to changes in implied volatility.
Rho (ρ)	Sensitivity to interest rate changes.

10.1.2 Basic Strategies

- **Covered Call** — own stock + sell call = generate income, cap upside
- **Protective Put** — own stock + buy put = insure downside
- **Straddle** — buy call + buy put at same strike = profit from big move either way
- **Vertical Spread** — buy one option + sell another at different strike = defined risk/reward

Options involve leverage and time decay. Short options have unlimited or very large risk. Understand all mechanics before trading with real capital.

10.2 Futures Contracts

A **futures contract** is a **standardised agreement** to buy or sell an asset at a predetermined price on a specific future date.

Uses:

- **Hedging** — lock in a price (e.g., airline hedging jet fuel)
- **Speculation** — directional bets with leverage
- **Arbitrage** — exploit price discrepancies between spot and futures

Key futures markets: **equity index** (S&P 500 /ES, NASDAQ /NQ), **bond** (US Treasury /ZB, /ZN), **energy** (Crude /CL, Natural Gas /NG), **metals** (Gold /GC, Silver /SI), **currencies** (/6E, /6J).

10.2.1 Contango and Backwardation

- **Contango**: Futures price > spot price. Normal for storage-cost commodities. Rolling long positions forward incurs a cost.
- **Backwardation**: Futures price < spot price. Indicates supply tightness. Rolling forward generates a positive roll yield.

10.3 Trading Term Cards (Finance Jargon)

10.3.1 Assets and Cash Markets

Key Term: Equities

Stocks / shares that represent ownership in a company. If the company grows, the equity value can rise; if it deteriorates, it can fall.
Example: Buying 1 share of a tech company makes you a partial owner of its future cash flows (via dividends and/or resale value).

Key Term: Fixed Income

Securities where returns are driven primarily by interest payments and credit quality (for example: government or corporate bonds). You earn coupons, and your total return depends on yields moving after you buy.
Example: Buy a 5% coupon bond; if market yields drop, the bond price can rise even if the coupon stays the same.

Key Term: FX (Foreign Exchange)

Trading currencies in pairs (e.g., **EUR/USD**). An FX quote is just the price of one currency in terms of another.
Example: If EUR/USD moves from 1.1000 to 1.1050, EUR strengthened versus USD.

Key Term: Commodities

Raw materials (oil, gold, wheat, copper, etc.) traded directly (spot) or via derivatives (futures/options). Many commodities are priced in USD, so FX and macro matter a lot.
Example: Oil prices can jump on supply disruptions; gold often responds to real rates and risk sentiment.

10.3.2 Derivatives

Key Term: Futures

Exchange-traded, standardised contracts to buy or sell an underlying at a future date for a predetermined price. Futures are marked-to-market (gains/losses flow daily).
Example: Hedge jet fuel costs for next quarter by locking in the fuel price via futures.

Key Term: Options

Contracts that give the holder the **right** (not obligation) to buy (call) or sell (put) the underlying at a **strike** price on or before expiration.
Example: A call option benefits if the stock rises above strike; your downside is limited to the premium you paid.

Key Term: Perp Swaps (Perpetual Futures Swaps)

Derivative contracts with no fixed expiration date ("perpetual"). Traders typically use them like futures, but the contract uses **funding** to keep the perp price close to the relevant spot/mark price.
Example: If BTC perp trades too high versus BTC spot, funding tends to push the perp back down by incentivizing more short demand.

Key Term: Funding Rate in Perp Swaps

A periodic payment exchanged between longs and shorts. The sign/magnitude is designed to reduce persistent price gaps (basis) between perp and spot.
Example: If perp price trades above spot (positive premium), longs typically pay shorts (funding is positive for longs), discouraging over-longing.

Key Term: Forwards

OTC (custom) contracts agreeing to buy/sell an asset at a future date for a fixed price. Forwards are not standardized like exchange futures and therefore typically carry more **counterparty risk**.
Example: A producer and airline might agree on fixed fuel pricing for next year via an OTC forward.

10.3.3 Position Direction and Portfolio Risk

Key Term: Long

Holding a position that benefits when the underlying price rises.
Example: Buying BTC at $60,000 and selling at $63,000 profits if your entry is lower than your exit.

Key Term: Short

Holding a position that benefits when the underlying price falls (via borrowing/selling or via derivatives like futures/perps).
Example: If you short at $100 and it falls to $95, you profit (before fees/interest).

Key Term: Flat

No active exposure: your net position is zero (or close enough that price moves do not meaningfully affect your P&L).
Example: After closing a long and a short, you are flat and stop taking directional risk.

Key Term: Exposure

How sensitive your portfolio is to a specific risk factor (directional price moves, rates, FX, volatility, etc.). Think of exposure as "how much you win/lose if the variable moves."
Example: Long 1 BTC means your mark-to-market moves roughly with BTC price by about 1:1 (ignoring funding/fees).

Key Term: Margin

Collateral you post to support leveraged positions. Margin determines how much loss you can absorb before a forced close.
Example: With $1,000 initial margin, you might control $10,000 notional (10x), but you only have $1,000 of buffer.

Key Term: Leverage

Controlling a larger notional position relative to your posted equity/collateral. Leverage amplifies both gains and losses.
Example: 10x leverage means a 1% adverse move can cost about 10% of your margin (ignoring fees and mark-price mechanics).

Key Term: Liquidation

The forced closing of a leveraged position when collateral/equity falls below the maintenance requirement (or the exchange's liquidation threshold).
Example: BTC drops quickly; your margin cushion is depleted and the exchange closes you at the prevailing mark/liquidation price.

Key Term: PnL (Profit and Loss)

The amount you gain or lose from a trade. It can be realized (closed positions) or unrealized (mark-to-market on open positions).
Example: Buy at $100 and sell at $110: realized P&L is +$10 per unit.

Key Term: Sharpe Ratio

A performance metric that rewards higher returns while penalizing volatility:

$$S = \frac{R_p - R_f}{\sigma_p}$$

where R_p = portfolio return, R_f = risk-free rate, and σ_p = return volatility.
Example: Two strategies both return 10%; if one is much smoother (lower volatility), it tends to have a higher Sharpe.

Key Term: Counterparty Risk

Risk that the other side in a contract fails to perform (e.g., defaults on payments). Exchange-cleared derivatives often reduce this risk compared to OTC contracts.
Example: An OTC forward depends on your counterparty paying/settling correctly.

10.3.4 Order Book, Execution, and Trading Costs

Key Term: Bid

The highest price a buyer is willing to pay.
Example: If the bid is $99.90, buyers are currently willing to buy at $99.90.

Key Term: Ask

The lowest price a seller is willing to accept.
Example: If the ask is $100.10, sellers are currently offering to sell at $100.10.

Key Term: Spread

The difference between ask and bid. In liquid markets, the spread is your baseline trading cost (you usually pay it when you enter/exit).
Example: Bid $99.90, ask $100.10: spread is $0.20.

Key Term: Fill

When an order is executed. Orders can fill partially (some quantity filled) or fully.
Example: Your $10,000 limit buy might fill $6,000 if remaining book liquidity doesn't reach your price.

Key Term: Slippage

The difference between the price you expected and the actual execution price, driven by spread, fast market moves, and your order size/impact.
Example: You expect a fill near $100 but the market moves and you end up filling at $100.25.

Key Term: Market Order

An order that executes immediately at the best available prices. You prioritize speed over price certainty.
Example: In a thin order book, a market buy may walk up the book.

Key Term: Limit Order

An order that executes only at your specified price or better. You prioritize price certainty over guaranteed fill.
Example: A buy limit at $99.50 won't buy if the ask never reaches $99.50.

Key Term: Tick

The smallest allowed change in price (minimum price increment).
Example: If tick size is 0.01 USDT on a venue, prices move in discrete steps of 0.01.

10.3.5 Market Makers, Inventory, and Crypto Basis

Key Term: Inventory

The net positions market makers accumulate while providing two-sided liquidity. Carrying inventory creates risk if the market moves against the position.
Example: If a market maker has sold more than bought recently, they may be net short and can lose if price rises.

Key Term: Skew

An asymmetry in pricing/quoting (often referring to options implied-volatility skew, or more generally that market makers price different strikes differently).
Example: Downside puts can have higher implied volatility, creating a skew across strikes.

Key Term: Spread Capture

How market makers try to profit: repeatedly buying at the bid and selling at the ask. If execution is mostly from passive flow, this can be a steady return stream.
Example: Earn small profits on many trades while keeping your quotes tight.

Key Term: Adverse Selection

The risk that trades hitting your quotes are informed (you get "picked off") rather than random. It can turn spread capture into losses.
Example: You quote tightly, but sophisticated traders know price is about to move and your filled trades become systematically loss-making.

Key Term: Basis

The price gap between two related instruments (commonly: futures/perps vs spot). In crypto, basis is tightly connected to funding.
Example: If BTC perp trades $200 above BTC spot, positive basis creates incentive for funding to move the perp back toward spot.

Key Term: Open Interest

The total number of outstanding derivative contracts (not yet closed). It reflects how much positioning is currently "hanging out" in the system.
Example: Open interest rising alongside price can indicate growing leverage; falling open interest can indicate deleveraging.

10.4 Quantitative and Algorithmic Trading

10.4.1 Systematic vs. Discretionary

- **Discretionary** — human makes all trading decisions. Can incorporate qualitative/contextual reasoning.
- **Systematic** — rules-based; computer executes all trades. Removes emotion. Requires robust backtesting and forward testing.

10.4.2 Strategy Development Process

1. **Hypothesis** — define the edge you believe exists and why it should persist.
2. **Data collection** — quality historical data is non-negotiable.
3. **Backtest** — simulate strategy on historical data. Watch for overfitting.
4. **Walk-forward test** — test on data not used in development (out-of-sample).
5. **Paper trading** — simulate live with real-time data, no real money.
6. **Live trading (small size)** — deploy with minimal capital first.
7. **Scale** — increase size only after demonstrable live performance.

Overfitting is the cardinal sin of backtesting: a strategy that perfectly fits historical data but fails on new data. The more parameters you optimise, the more you risk fitting noise. Always hold out out-of-sample data for validation.

10.4.3 Common Quant Strategies

- **Momentum** — buy winners, short losers over 3–12 month lookbacks
- **Mean reversion** — bet on extreme deviations reverting to the mean
- **Statistical arbitrage (stat-arb)** — pairs or basket mean reversion
- **Trend following** — CTA style; long trending assets across all classes
- **Factor investing** — exposure to value, quality, size, low-volatility factors

10.5 Behavioural Finance

Key Term: Behavioural Finance

The study of how cognitive biases and emotional responses cause investors to make irrational decisions, leading to systematic market mispricings.

Bias	Description
Loss Aversion	Losses feel roughly 2× worse than equivalent gains feel good
Overconfidence	Overestimating one's predictive accuracy and skill
Anchoring	Over-relying on the first piece of information seen (e.g., purchase price)
Recency Bias	Overweighting recent events; extrapolating the current trend
Herd Behaviour	Following the crowd; contributing to bubbles and crashes
Disposition Effect	Holding losers too long, selling winners too early
Sunk Cost Fallacy	Holding a bad investment to "get back to even"
Narrative Fallacy	Constructing plausible stories to explain random price moves

10.6 Recommended Learning Roadmap

10.6.1 Beginner (Months 1–3)

- Complete this entire notes document
- Read: *The Intelligent Investor* — Benjamin Graham
- Read: *A Random Walk Down Wall Street* — Burton Malkiel
- Open a **paper trading account** (Interactive Brokers, Thinkorswim)
- Study and annotate 100 historical charts

10.6.2 Intermediate (Months 4–9)

- Read: *Market Wizards* — Jack Schwager
- Read: *Trading in the Zone* — Mark Douglas (psychology)
- Read: *Technical Analysis of the Financial Markets* — John Murphy
- Develop and paper trade a *single, well-defined* strategy for 3 months minimum
- Begin tracking trades in a **trading journal**

10.6.3 Advanced (Month 10+)

- Trade live, minimum size; focus on process, not P&L
- Read: *Reminiscences of a Stock Operator* — Edwin Lefèvre
- Study options via tastytrade or Options Education (CBOE)
- Explore macro with: *The Alchemy of Finance* — George Soros
- Consider Python for backtesting: `pandas`, `vectorbt`, `backtrader`

10.7 Essential Tools and Resources

Tool	Category	Use Case
TradingView	Charting	Best charting platform; scripting via Pine Script
Interactive Brokers	Broker	Best for serious traders; global access, low fees
Bloomberg Terminal	Data/News	Institutional standard (expensive); use free alternatives
Yahoo Finance	Data	Free fundamental data and price history
EDGAR (SEC)	Filings	Official source for US company filings
Finviz	Screener	Stock screening by technical and fundamental filters
Quandl / FRED	Economic Data	Macro and economic time series data
Python / R	Quant	Backtesting, data analysis, strategy automation

10.8 Final Principles to Internalise

I. **Capital preservation first**. You cannot profit if you blow up.

II. **Process over outcome**. A losing trade within your rules is fine. A winning trade outside your rules is dangerous.

III. **The market is always right**. Your opinion is irrelevant; price is fact.

IV. **Cut losers, let winners run**. This is the opposite of what emotions dictate.

V. **There is no holy grail**. Any edge erodes over time; keep learning.

VI. **Consistency beats brilliance**. A 60% win rate with 2:1 RRR, repeated 1,000 times, compounds to large wealth.

VII. **Know your edge**. If you cannot articulate why your strategy works, you do not have one.

VIII. **Keep a trading journal**. The data in your own trade history is the most valuable dataset you will ever have.

> **"The goal of a successful trader is to make the best trades. Money is secondary."** — Alexander Elder